

CTP406 | Final Project

Mutant

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Brief

This project is an interactive work that utilizes two projectors. On one screen, an AI-generated face palette of 49 faces across the gender spectrum of a single person is displayed, while on the left, a shadowy figure moves behind a marquee. As the audience clicks on specific faces with the mouse, the figure behind the marquee interacts with audio and visual. This work is an attempt to question and dismantle gender binaries. The work invites the audience to rethink the act of gender passing (clicking).



1. Background & Motivation

What is it like to live in South Korea as a queer person?
Through our work, we question the concept of gender binaries and invites the audience to rethink the act of gender passing.

1.1 Before team organization

Junyeong

He has a personal interest in the topic of ‘queer’.
He didn't feel ready to make work specifically about queer people yet.
He don't think the queer experience itself is special.

Dana

She has experience in private documentary production.
She is interested in the bias of AI and the distortion of truth.
She kept thinking about where is truth in documentaries.

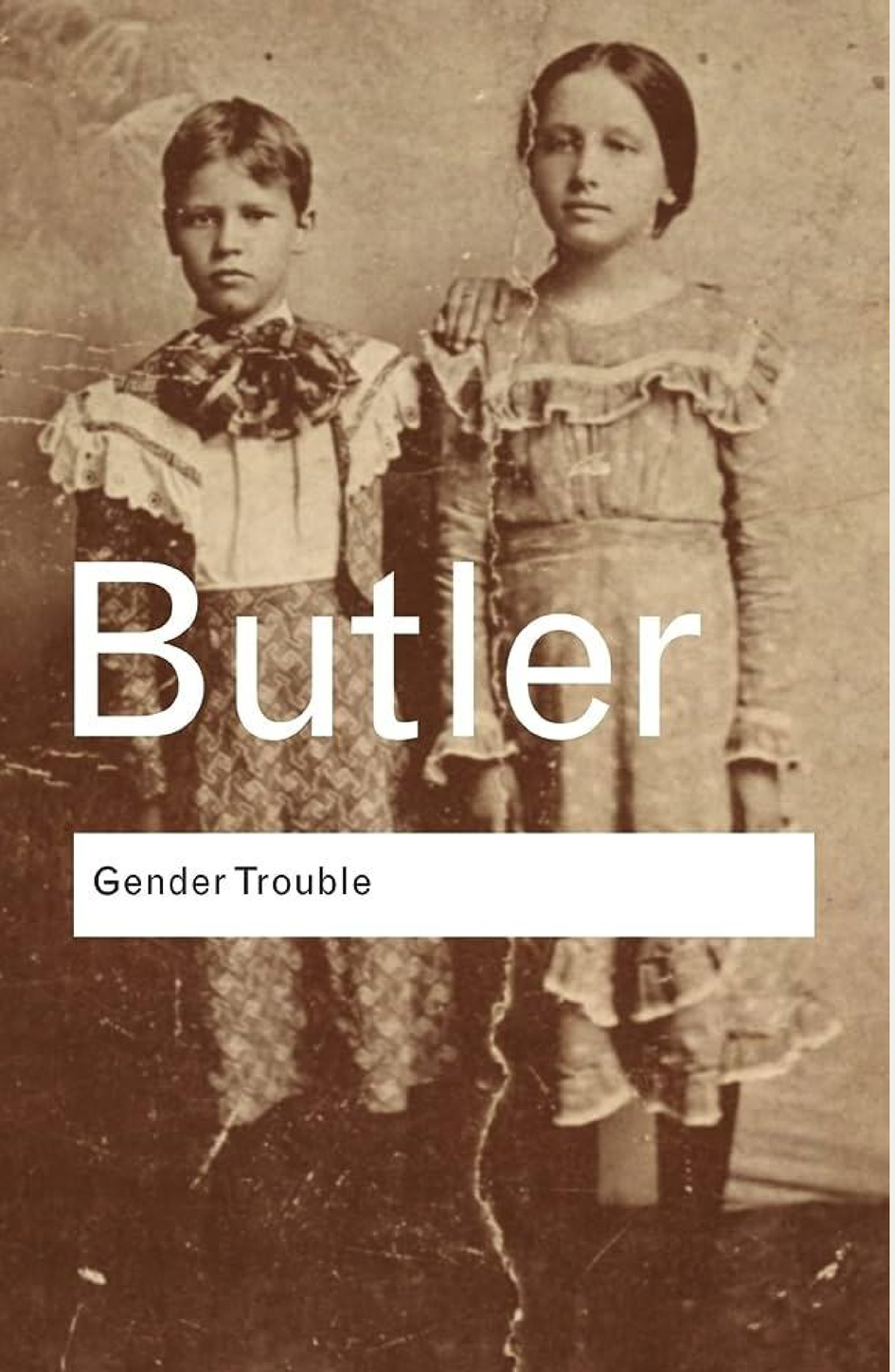
1.2 After team organization

Refine your topic through interview

“I think what I've always been interested in is I've always wanted to talk about subverting the gender binary... and I think I hate institutional binaries so much.” -Junyeong

Discussion on how to represent the topic

1. Representing parodic identities in the form of actual physical layers
2. Passing the process of performing drag through AI
3. Creating gender-neutral AI images
4. Transcending the concept of gender binaries through institutional subversion (queer people filling marriage licenses)
5. Revealing the contradictions of gender binaries through 'face' images → **adoption**



2. Literature Review

2.1 Gender

<Gender Trouble> by Judith Butler

[3] [2]

Gender Performativity

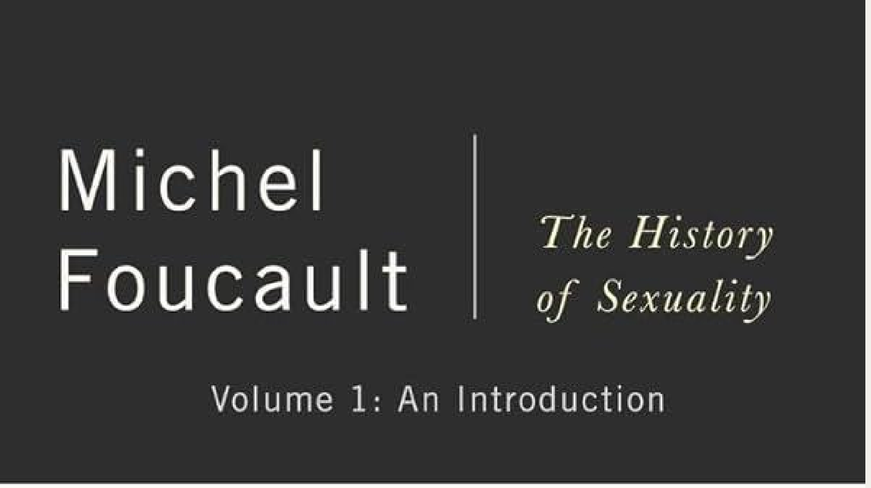
Butler defines gender as the result of repeated actions that produce the effect of a fixed or normal gender while simultaneously blurring the contradictions and instability of any individual's gendered behavior. These outcomes produce what we might recognize as "true gender," and "true gender" is a narrative sustained by an unspoken collective agreement to perform, produce, and maintain distinct and bipolar genders.

In Butler's hypothesis, the socially constructed aspects of gender performativity are most explicit in the performance of drag, which provides a fundamental understanding of gender binary. The exaggerated femininity and masculinity of drag reveals that what we think of as natural is a performance, an act. If "normalcy" is constructed through repeated human behavior, this means that "normalcy" can be subverted by performing different behaviors.

Parodic identity. Gender parody

A parodic identity is an identity in the sense of an originless imitation, as in a camouflage, or a masquerade, not an imitation of the original but an imitation of the imitation.

Gender parody stylizes gender itself and then reveals that the original identity is itself an originless imitation. In this sense of imitation without an origin, the parody is already linked to Baudrillard's simulation, which blurs the boundaries between original and copy. Butler draws on Hutcheon's parody and Baudrillard's simulacra to argue that the very act of imitation undermines the authenticity or authority of the original, so that the dichotomy of original/imitation is no longer possible.





2.1.1 Gender-Body

Difference and Repetition
"Repetition changes nothing in the object repeated, but does change something in the mind which contemplates it."
-Gilles Deleuze. (1968). Difference and Repetition

Phenomenology of the Body
"The body is the vehicle of being in the world, and having a body is, for a living creature, to be intervolved in a definite environment, to identify oneself with certain projects and be continually committed to them."
- Maurice Merleau-Ponty. (1945). Phenomenology of Perception

Body and Power
"Sexuality must not be thought of as a kind of natural given which power tries to hold in check, or as an obscure domain which knowledge tries gradually to uncover. It is the name that can be given to a historical construct."
- Michel Foucault. (1976). The History of Sexuality

The Gaze of the Other
"I become an object in the world of the other."
- Jean Paul Sartre. (1943). Being and Nothingness

Representation and Image
A representation is just a moment M of reality. Just as any point in space is a reflection of space y and a memory of time x, so every image is a moment. An image that does not contain any creation, any 'possibility of life' is a dead image.
Nicolas Burillo. (1998). Relational Aesthetics

Have I ever detected my own face?

Meaning of the face
"The face of the other reveals an infinity that cannot be contained within my possession or control."
-Emmanuel Levinas. (1961). Totality and Infinity

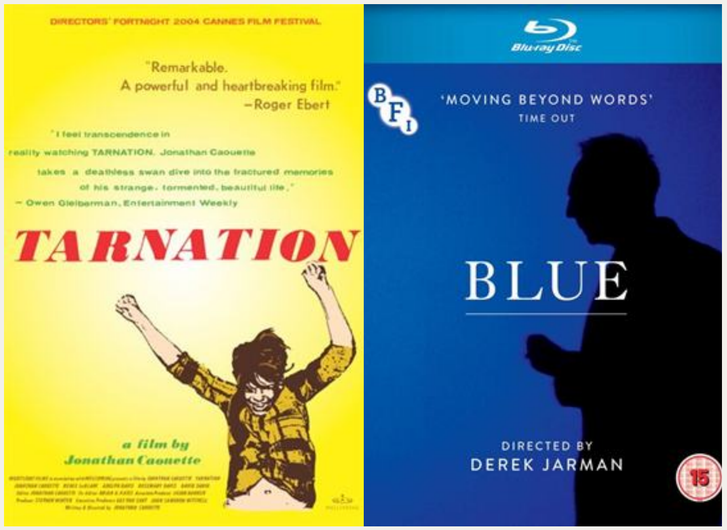
The Gaze of the Other
"I become an object in the world of the other."
- Jean Paul Sartre. (1943). Being and Nothingness

2.2 Documentary

Truth in Documentary

It is not the clarity of the documentary image, but the excited uncertainty that gives it the paradoxical power to dominate people. At this point, the point of documentary truth shifts. This is because it is no longer just a matter of whether documentary images match reality or not. Expression instead of expression. The video content may or may not match reality. But its form will inevitably tell the truth, the truth about the context of the video itself and its production and conditions.

- Hito Steyerl. (2008). The Colour of Truth



2.3 AI Technology

Mean Images

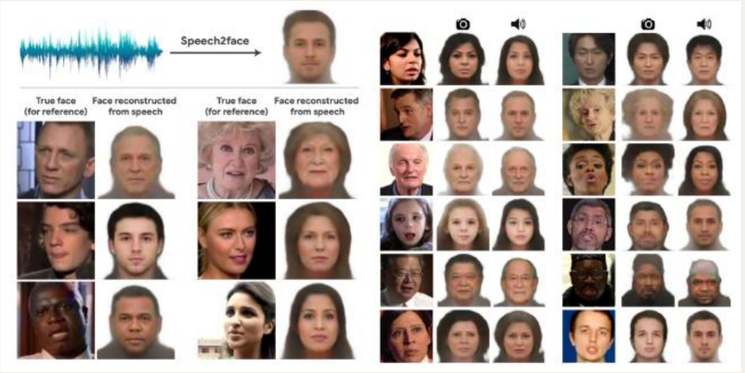
Visuals created by ML tools are statistical renderings, rather than images of actually existing objects. (...) The question is, what mean? Whose mean? Which one? Stable Diffusion renders this portrait of me in a state of frozen age range, produced by internal, unknown processes, spuriously related to the training data

- Hito Steyerl. (2023). Mean Images

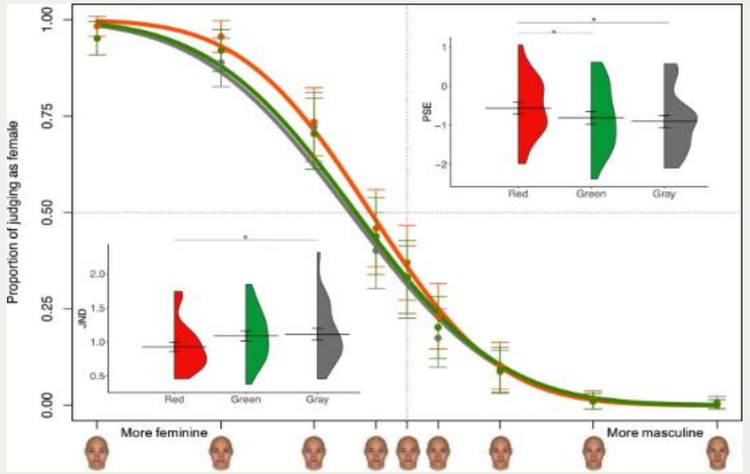
How does a computer 'see' gender?

"To better understand how this works, we showed images of the Center's staff members to a trained machine vision system similar to the one we used to classify image searches. We then systematically obscured sections of each image to see which parts of the face caused the system to change its decision about the gender of the person pictured."

- Pew research center. (2019). The challenges of using machine learning to identify gender in images



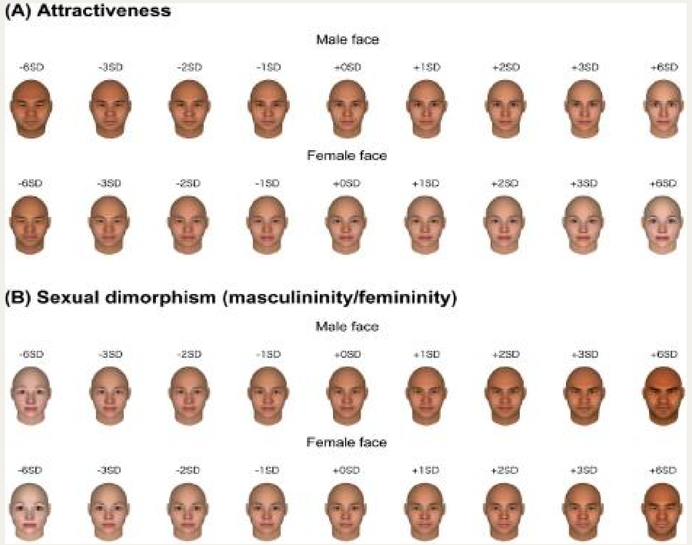
Speech2Face: Learning the Face Behind a Voice (2019)



Red background color biases gender categorization of human faces (2023)

2.3 Related Artwork

2.3.1 Face



Sexual dimorphism & Attractiveness (2020)



Passport don’ts (2017)



Andy Warhol, Ladies and Gentlemen (1975)



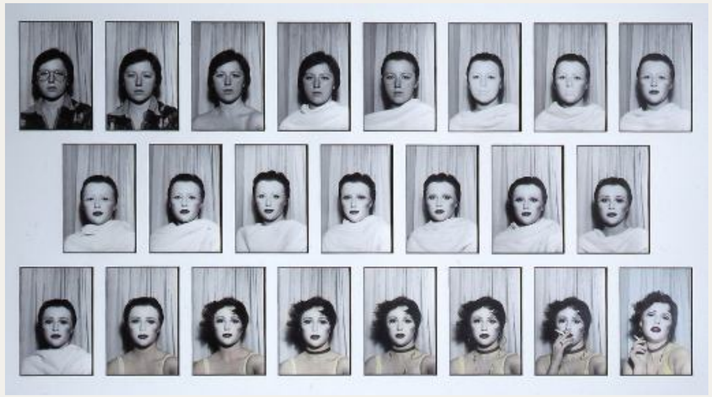
Marlene Dumas, Chlorosis (1994)



Byron Kim, Synecdoche (1991-)



ClauCahun, Que me veux tu? (1929)



Cindy Sherman, Untitled#479 (1975)

2.3.2 Repetition



Andy Warhol, Marilyn Monroe



Claude Monet, Rouen Cathedral Series (1892-94)

2.3.3 Gender

Learning about Butler's gender performativity led me to revisit Lee Bul’s work <Monster>.
Appearing in 1998, Lee Bul's <Monster> is an unprecedented creature born from the artist's fingertips. Resembling alien creatures, animal organs, and tender plant roots (as the artist asserts, "It's a monster because it transcends boundaries"), it is a metaphor for a subversive woman who has broken down the boundaries of animal-plant, nature-artificial, male-female, and has become othered herself. The same year, she created another monster, the cyborg. The cyborg, as discoursed by Donna Haraway, is a genderless 'postgender' that undermines patriarchy and the symbolic system.



Lee Bul <Monster>

3. Implementation
3.1 Process

Problem

Q. What kind of story do ‘I’ want to tell?
A. Stories that subvert the gender binary.

Q. How do you want to express it?
A. By connecting it to institutional limitations such as marriage, adoption, etc.

Can the message be communicated to non-queer people?

Solution

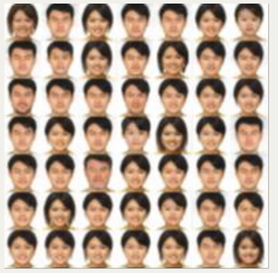
Listened to the feedbacks and made changes to make it fun and witty for non-queer audiences.

Based on research

Gender Parody and Gender Performativity
Difference and Repetition
Agential Realism in Gender
Totality and Infinity

Implementation

- The spectrum of gender is represented by a palette of 49 faces
- But the originality of gender is not in that palette - not anywhere
- 'My' performatively changing identity is represented as a shadow behind a veil



3. Implementation
3.1 Process

Q. What are the facts here?
A.
'I' am queer
'I' was recently told in class that 'queer people are mutants'
I want to break the mold of the gender binary.

Q. How do we communicate the facts?
A. By connecting it to institutional limitations such as marriage, adoption, etc.

Problem

When the “me” behind the curtain is a real interview
Doesn't fit the concept of “fun” and “wit
You can't be sure that the interview is the 'truth'

Solution

Turn my interview behind the curtain into a real-time changing dance. My continuous movement itself does not change based on the audience's clicks (color, shape, voice interaction)

Based on research

Truth in Documentary
Documentary and representation of reality
Body and Power
Phenomenology of the Body

→

Implementation

- Considering audience facial 'clicks' as specific 'gender passing'
- ‘My’ appearance performatively changes behind the curtain regardless of the audience's 'gender passing'
- The mood of 'me' is expressed in the interaction when the audience clicks
- Conveying facts through 'expression' rather than 'explanation'



3. Implementation
3.1 Process

Q. Q. To what extent can
the AI-derived me be called 'me'?

Problem

Can an AI-generated image reflect your identity?
None of the images feel like ‘me’

Solution

Titled “Mutant” based on feedback
AI's gender spectrum is not original
Rather, like a mutation, ‘I’ can find ‘myself’ changing in real time

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Based on research

Mean Images
The challenges of using machine learning to identify gender in images

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Implementation

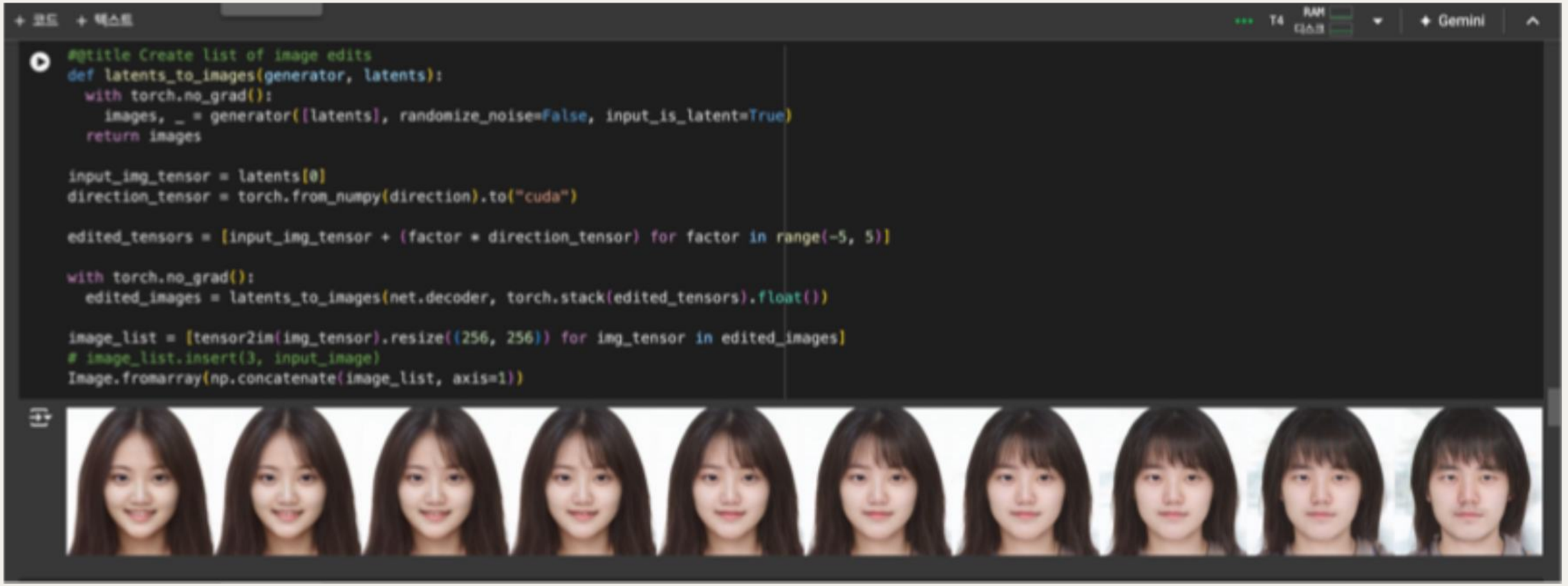
- Added an age variable to the face palette to extend the past-present-future spectrum and increase randomness
- But paradoxically emphasizes that there is no originality anywhere



3. Implementation
3.2 Methods

Adjusting facial features using StyleGAN : Adding Feminity and Masculinity

Additional synthetic data is extracted based on photos of each spectrum generated by AI

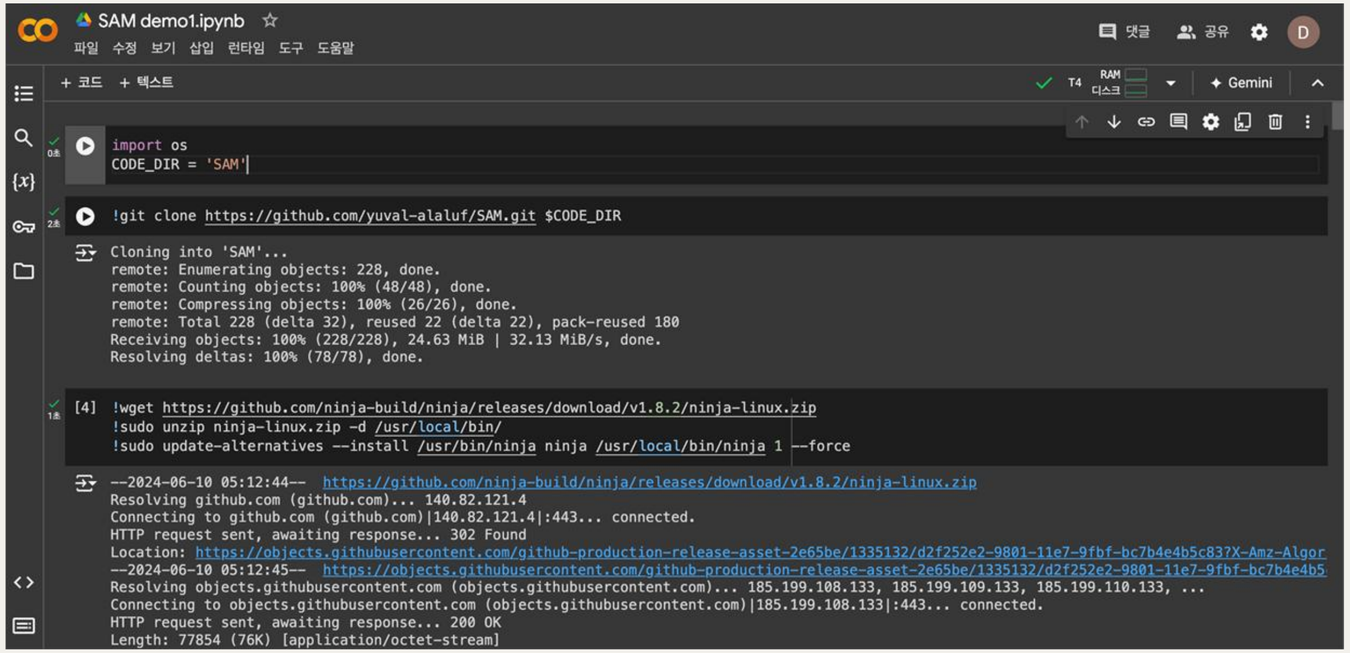


3. Implementation
3.2 Methods

Adjusting facial features using StyleGAN : Age variation

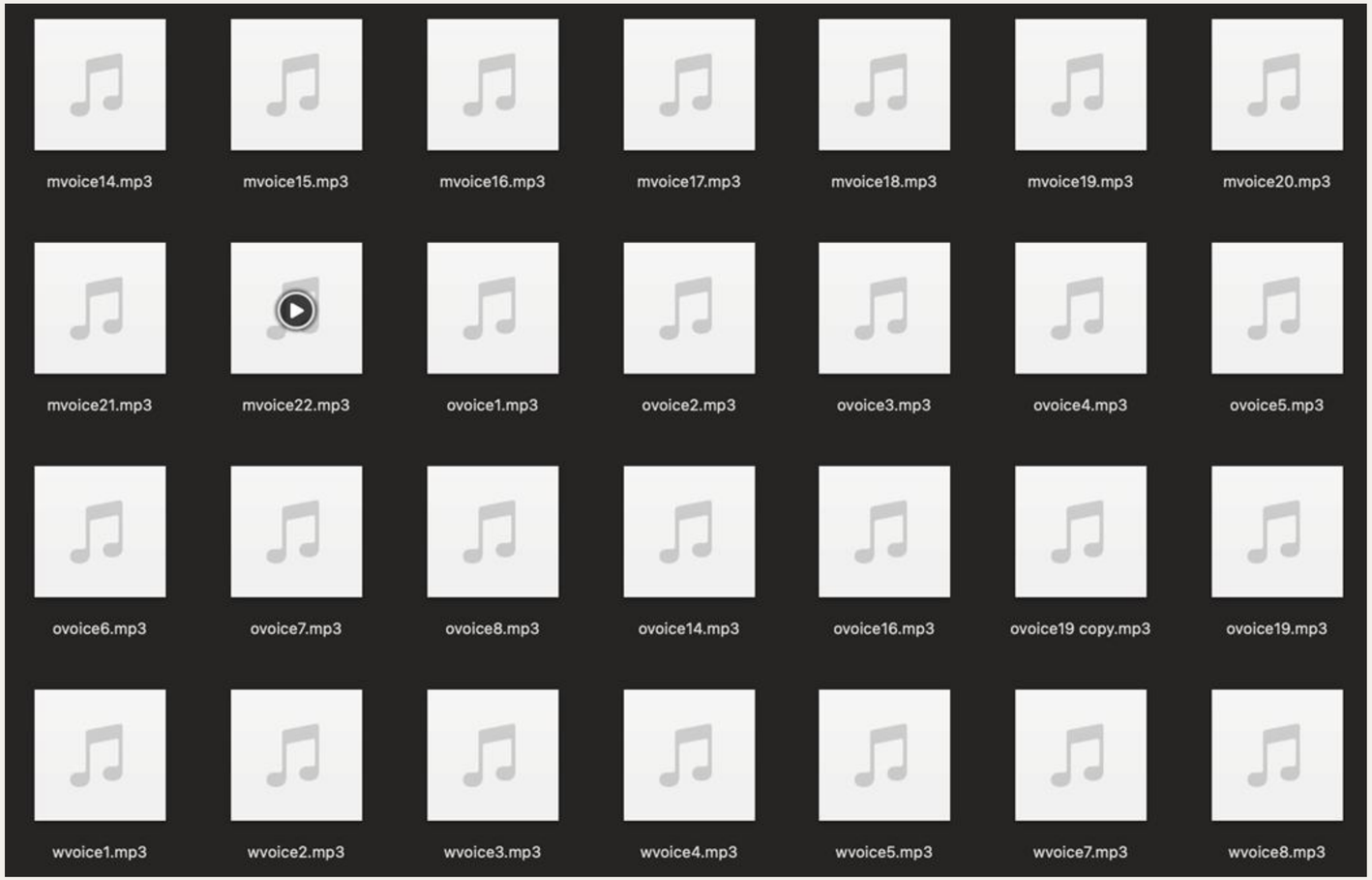
Additional synthetic data extracted based on photos within each face palette location generated by the AI





3. Implementation
3.2 Methods

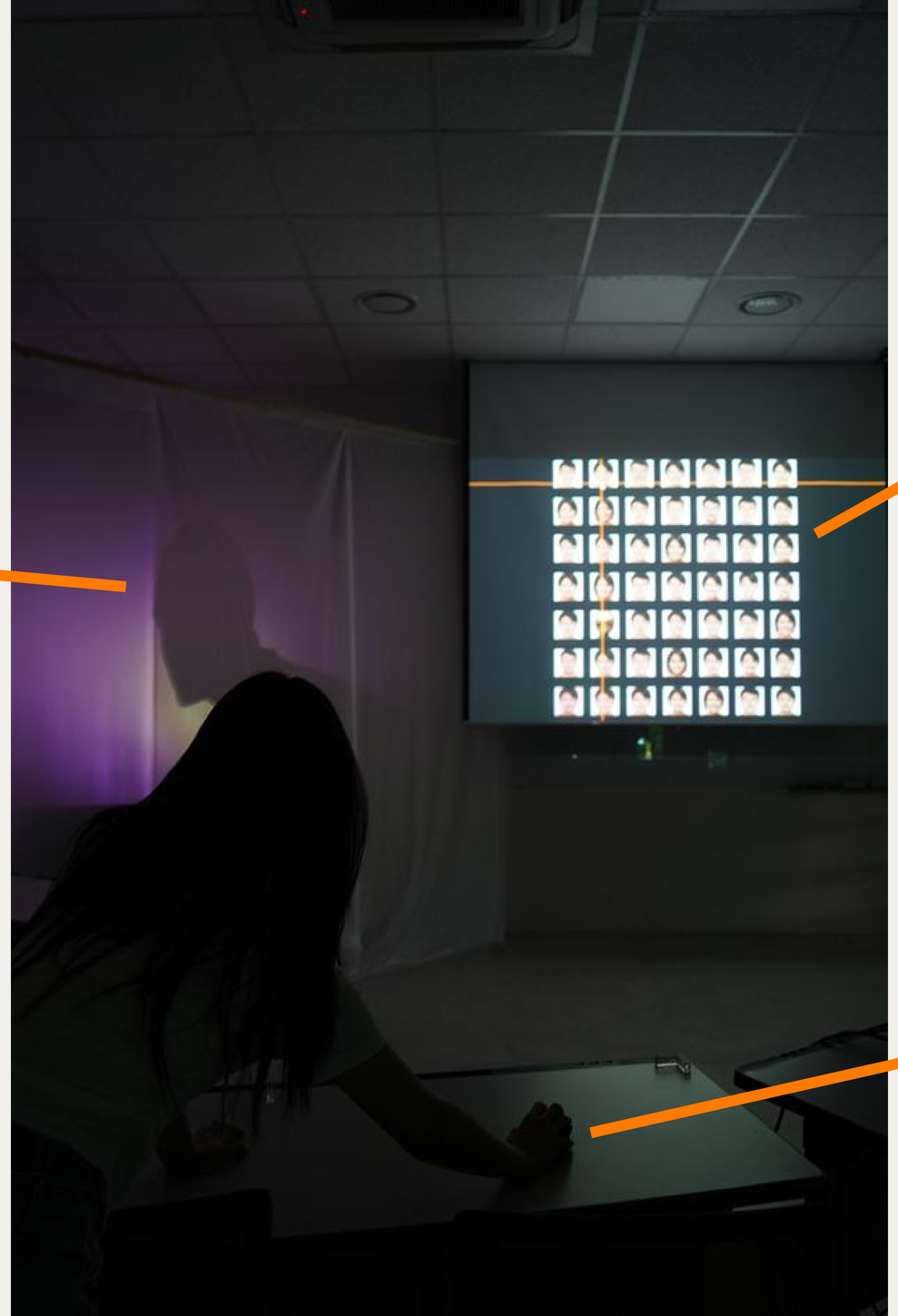
Implement 49 randomized voices Colab, Voxal App, Premiere



Reference Voice morphing

https://github.com/nestyme/voice-morphing/blob/7be5efa9201db43893e59a114e890608b7d0b2f0/demo.ipynb

3. Implementation
3.2 Methods



Face Palette
Me as multiple gender possibilities, created by AI (not real)
*added one ‘alien’ face to convey the message

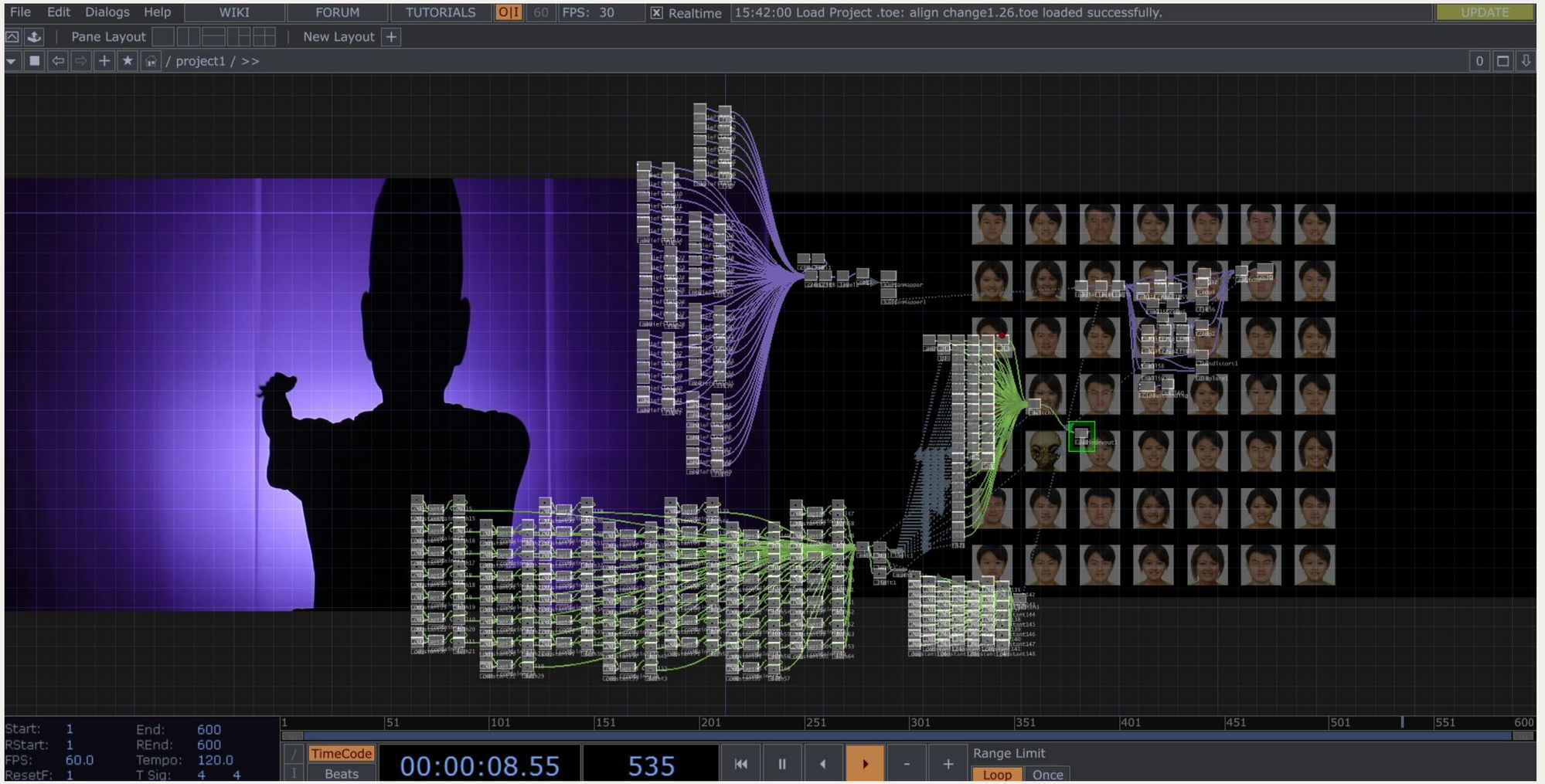
Shadow
My identity that evolves performance-wise, regardless of clicking

Sound
What I want to do (between laughing and crying) when I hear gender-binary sentences

Clicking
Gender passing by audiences

3. Implementation
3.2 Methods

Using **TouchDesigner** for Implementation





4. Result
4.1 Feedbacks - Our Team

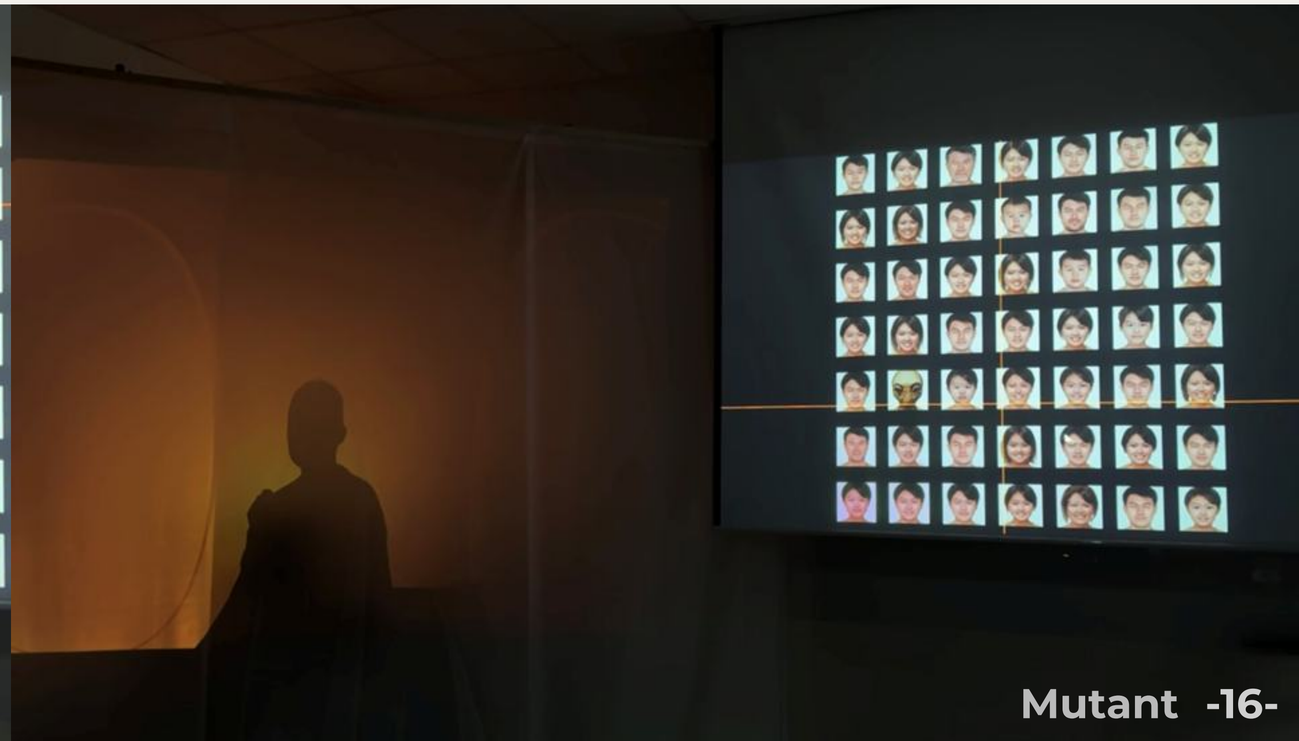
2)

- After extracting the gender spectrum based on the original face data, we additionally extracted synthetic data with gender/age variables to randomize the palette to better convey the intention.

- The visual representation behind the curtain that responded to the audience's passing was appropriate.

- It would have been better if the voice was also AI-based and precise.





4.2 Feedbacks - Others

“It's cool and fun to see it all on one screen”[2)]
“It's kind of scary and exciting at the same time”
“What happens when you click on this alien?”
“I didn't realize it was a picture of one person!”







5. Discusstion & Conclusion

5.1 Contribution Point

- Researched 'future art' and explored new formats to capture personal stories beyond traditional video shooting
- Expanded the scope of facial artwork to include the keywords 'queer' and 'gender' using AI.
- Created opportunities for non-queer groups to actively participate in the project by making a witty concept around the theme of ‘mutation’

5.2 Limitation and Future Work

“One person commented on the face palette and said that even the photos that were eventually transformed were identified as male or female, but ‘I’ think what ‘I’ was really trying to say was that all of those palettes shouldn't pass for any gender. ‘I’ think the gender binary is still so ingrained in so many people that it's hard to think outside of it.”

- Didn't achieve what we initially set out to do with the documentary form of film.
- In the future, we would like to expand the ways in which we can incorporate future art with the theme of gender based on the genre of documentary.

Reference

Theoretical Background

Judith Butler. (1990). Gender Trouble: Feminism and the Subversion of Identity
Karen Barad. (2007). Meeting the Universe Halfway: Quantum Physics and the Entanglement of Matter and Meaning
Maurice Merleau-Ponty. (1945). Phenomenology of Perception
Michel Foucault. (1976). The History of Sexuality
Gilles Deleuze. (1968). Difference and Repetition
Nicolas Burillo. (1998). Relational Aesthetics
Emmanuel Levinas. (1961). Totality and Infinity
Jean Paul Sartre. (1943). Being and Nothingness
Hito Steyerl. (2008). The Colour of Truth
Hito Steyerl. (2023). Mean Images
Pierre Leby. (1997). Cyberculture
Choi Hyunju. (2018). Documentary and representation of reality

Artworks

Derek Jarman, Blue (1993)
Jonathan Caouette, Tarnation (2003)
Lee Bul, Monster (1998-2001)
Ian Cheng,Bob(Bag of Beliefs) (2018-2019)
Claude Cahun, Que me veux tu? (1929)
Marlene Dumas, Chlorosis (1994)
Byron Kim, Synecdoche(1991-)
Claude Monet, Rouen Cathedral Series (1892-94)

Technical Research

Nikita Shiledarbaxi. (2021). Hands-on Python Guide to Style-based Age Manipulation (SAM) Technique
Tae-Hyun Oh, Tali Dekel, Changil Kim, Inbar Mosseri, William T. Freeman, Michael Rubinstein, Wojciech Matusik. (2019).Speech2Face: Learning the Face Behind a Voice
Na Chen, Koyo Nakamura & Katsumi Watanabe. (2023).Red background color biases gender categorization of human faces
Koyo Nakamura & Katsumi Watanabe.(2020). A new data-driven mathematical model dissociates attractiveness from sexual dimorphism of human faces
Steven Heller. (2019). Facial Recognition

Technical Application

Isabel Hermes. (2022). Utilizing StyleGAN to Explore Gender Directions in Face Image Editing
YUVAL ALALUF, PATASHNIKDANIEL COHEN. (2021). Only a Matter of Style: Age Transformation Using a Style-Based Regression Model